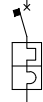
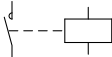
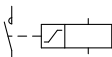
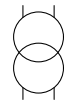


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TENDRARA PROJECT GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS																									
SAFETY CRITICAL ELEMENT LIST																									
IFI	2	31/01/2025	AS-BUILT				ADF	GDP	IC																
IFI	1	06/12/2024	GENEERAL REVIEW				ADF	GDP	IC																
IFI	0	26/03/2024	ISSUED FOR COMMENTs				ADF	GDP	IC																
Status	Rev. n.	Rev. n.	Rev. n.				Rev. n.	Rev. n.	Rev. n.																
	Rev. Index	Date	Description				Prepared	Checked	Approved																
Client							Project Title																		
ManaEnergy Managem Group MANAENERGY Mahaj Riad Center, Batiment 6, 3eme etage Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco							TENDRARA PROJECT																		
Contractor							Provision of a m-LNG plant for production for ManaEnergy																		
ITALFLUID ITALFLUID GEOENERGY Marocco Branch Boulevard de Bourgogne Rue Jaafar Ibn Habib Residence Al Mashrik II 1ere Etage N3 Casablanca																									
Engineering																									
ITALFLUID ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) Tel:085-4224374 / fax: 085-4220742																									
Facility Name			Location				Scale		Sheet of Sheets																
TENDRARA			MOROCCO				A4		1 of 18																
Document Title					Doc. Code																				
P-PK-PK-341-2					ELD01-ING-FR01																				
AMMONIA COMPRESSOR PACKAGE 2					Vendor Doc. code																				
					IE24090_SE08_P-PK-PK-341-2																				
			Project : P-PK-PK-341-2				Customer : ManaEnergy Managem Group				Object :														
General Contractor :		Manufacture : esa esaelettromeccanica.com		AMMONIA COMPRESSOR PACKAGE 2				Contract : S/N :				IE24090 IE24090_SE06		Rev. 2 Rev. 1				Data : 31/01/2025 Data : 06/12/2024				Date : 26/03/2024		Sheet : 1 of : 18	

CIRCUIT SYMBOLS ACCORDING TO CEI STANDARD

	ISOLATOR		ELECTRONIC TIMER WITH PICK-UP DELAY		MUSHROOM PUSH-BUTTON PULSE CONTACT WITH STAY-PUT MAKE CONTACT OPERATED BY STRIKING		STEDY OR FLASHING LIGHT
	FUSED ISOLATOR		ELECTRIC PULSE METER		KEY OPERATING DEVICE FOR SELECTORS, PUSH-BUTTON, ETC.		PLC UNIT
	CIRCUIT BREAKER WITH THERMAL AED MAGNETIC OVERCURRENT RELEASES		FUSE CARRIER		PRESSURE SWITCH		CONTROL INSTRUMENT
	CIRCUIT BREAKER WITH MAGNETIC RELEASE		MAKE CONTACT OF CONTACTOR		TEMPERATURE SWITCH		ENERGY METER
	CIRCUIT BREAKER WITH THERMAL, MAGNETIC AND DIFFERENTIAL RELEASES		MAKE CONTACT OF CONTACTOR		FLOW SWITCH		MEASURING INSTRUMENT
	RESIDUAL CURRENT CIRCUIT BREAKER "AC" TYPE FOR ALTERNATING CURRENT "A" TYPE FOR ALTERNATING OR PULSATING CURRENTS		MAKE AND BREAK CONTACT		FLOAT SWITCH		VOLTAGE TRANSFORMER
	THERMAL RELAY (OVERLOAD RELAY)		MAKE AND BREAK CONTACT WITH DROP-OUT DELAY		BUCHHOLZ RELAY		CURRENT TRANSFORMER
	COIL OF CONTACTOR RELAY		MAKE AND BREAK CONTACT WITH PICK-UP DELAY		CAM OPERATED DEVICE		ACOUSTIC SIGNALLING DEVICE
	COIL OF SHUT-OFF ALARM MAKE CONTACT ON ENERGIZATION		MAKE AND BREAK CONTACT WITH DROP-OUT AND PICK-UP DELAY (FLIP-FLOP)		PROXIMITY SWITCH		RESISTANCE
	COIL OF SHUT-OFF ALARM DENERGIZES IN CASE OF UNDERVOLTAGE		MAKE AND BREAK CONTACT OF THERMAL RELAY		TEMPERATURE SENSOR		CAPACITOR
	COIL OF OPERATING DEVICE WITH DROP-OUT DELAY		MAKE AND BREAK CONTACT OF LIMIT SWITCH		PHOTOELECTRIC SENSOR		SINGLE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH PICK-UP DELAY		MULTI POSITION SWITCH (GENERAL SYMBOL)		SOLENOID VALVE		THREE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH DROP-OUT AND PICK-UP DELAYS		MAKE AND BREAK CONTACT OPERATED BY TURNING WITH SPRING RETURN		PHOTOCELL		
	COIL OF LATCHING RELAY (STEP-BY-STEP FUNCTION)		MAKE AND BREAK CONTACT OPERATED BY PUSHING WITH SPRING RETURN				
	ELECTRONIC OR ELECTRO-MECHANICAL TIME SWITCH		MAKE AND BREAK CONTACT OPERATED BY PULLING WITH OR WITHUOT SPRING RETURN				

CONDUCTOR COLOR CODING INSIDE CONTROL PANEL

AC AND DC POWER CIRCUIT	BLACK
NEUTRAL CONDUCTOR OF POWER CIRCUIT	LIGHT-BLUE
PROTECTIVE CONDUCTOR (EARTH CONDUCTOR)	YELLOW-GREEN
110/220V AC AUXILIARY CIRCUIT (*)	RED
24V AC AUXILIARY CIRCUIT	WHITE
24V AC AND DC PLC INPUT	GRAY
24V AC PLC OUTPUT	WHITE
24V AC AND DC INTERLOCKING CIRCUIT	ORANGE
24V DC AUXILIARY CIRCUIT AND PLC OUTPUT	DARK-BLUE
INTRINSIC SAFE SIGNAL	BLUE

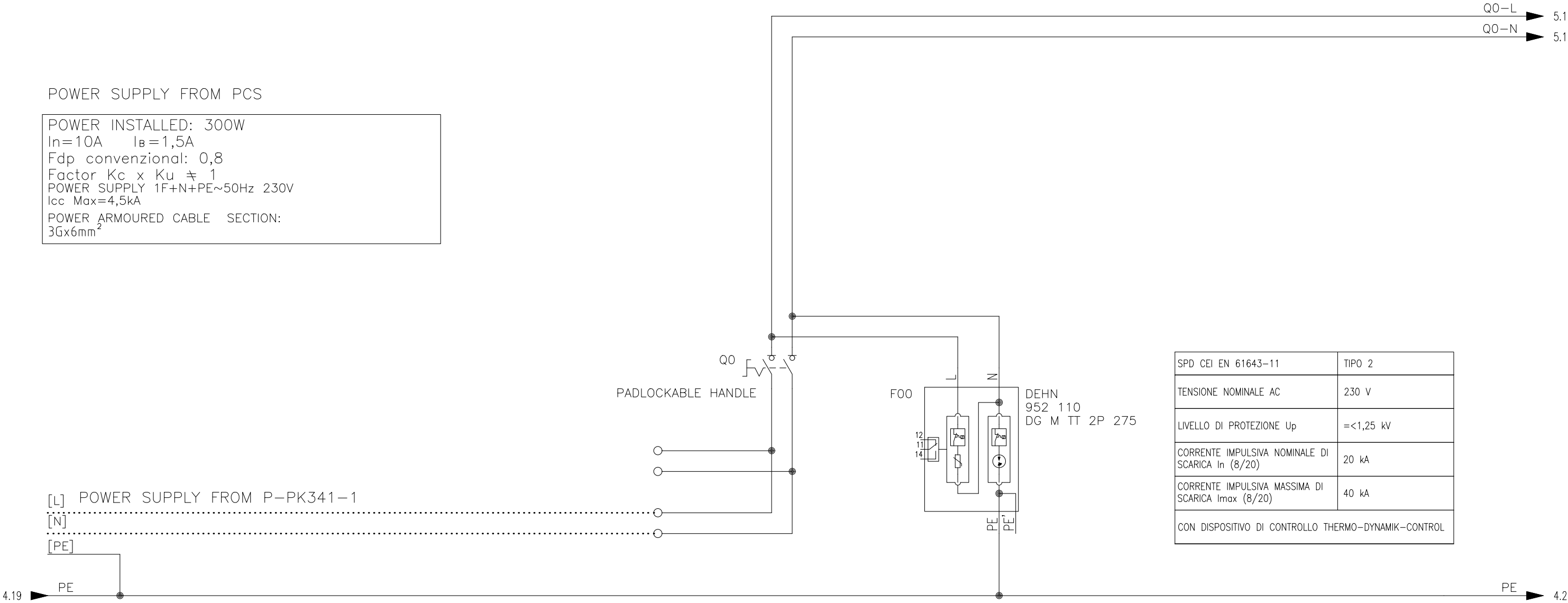
(*) : RED FOR 110V AND BROWN FOR 220V, IN CASE BOTH VOLTAGE ARE USED

CODE COLOUR

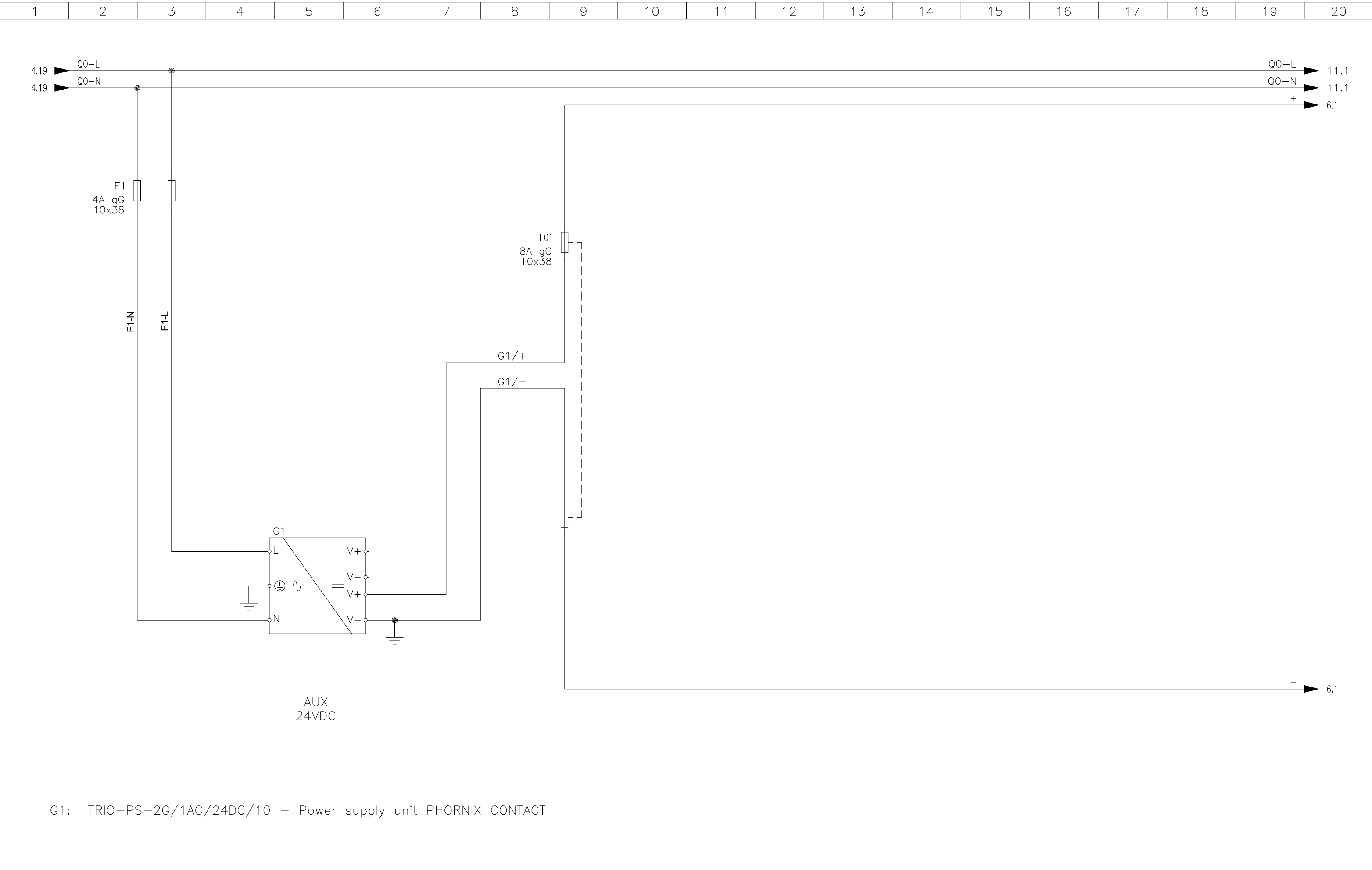
COLOUR	CODE LETTERS
BLACK	BK
BROWN	BN
RED	RD
ORANGE	OG
YELLOW	YE
GREEN	GN
BLUE	BU
GREY	GY
WHITE	WH
SILVER	SR
CLEAR	CL
GREEN AND YELLOW	GNYE

		Project : P-PK-PK-341-2 AMMONIA COMPRESSOR PACKAGE 2	Customer : 	Object :			
				By :			
General Contractor :	Manufacture : 		Contract : IE24090	Rev. 2	Data : 31/01/2025	Date : 26/03/2024	
			S/N : IE24090_SE06	Rev. 1	Data : 06/12/2024	Sheet : 3	of : 18

	POWER MAIN DISCONNECT SWITCH																FEEDER ID		
	1F+N+PE~50Hz 230V																RATED POWER Pn-RATED VOLTAGE Vn		
																	RATED CURRENT		

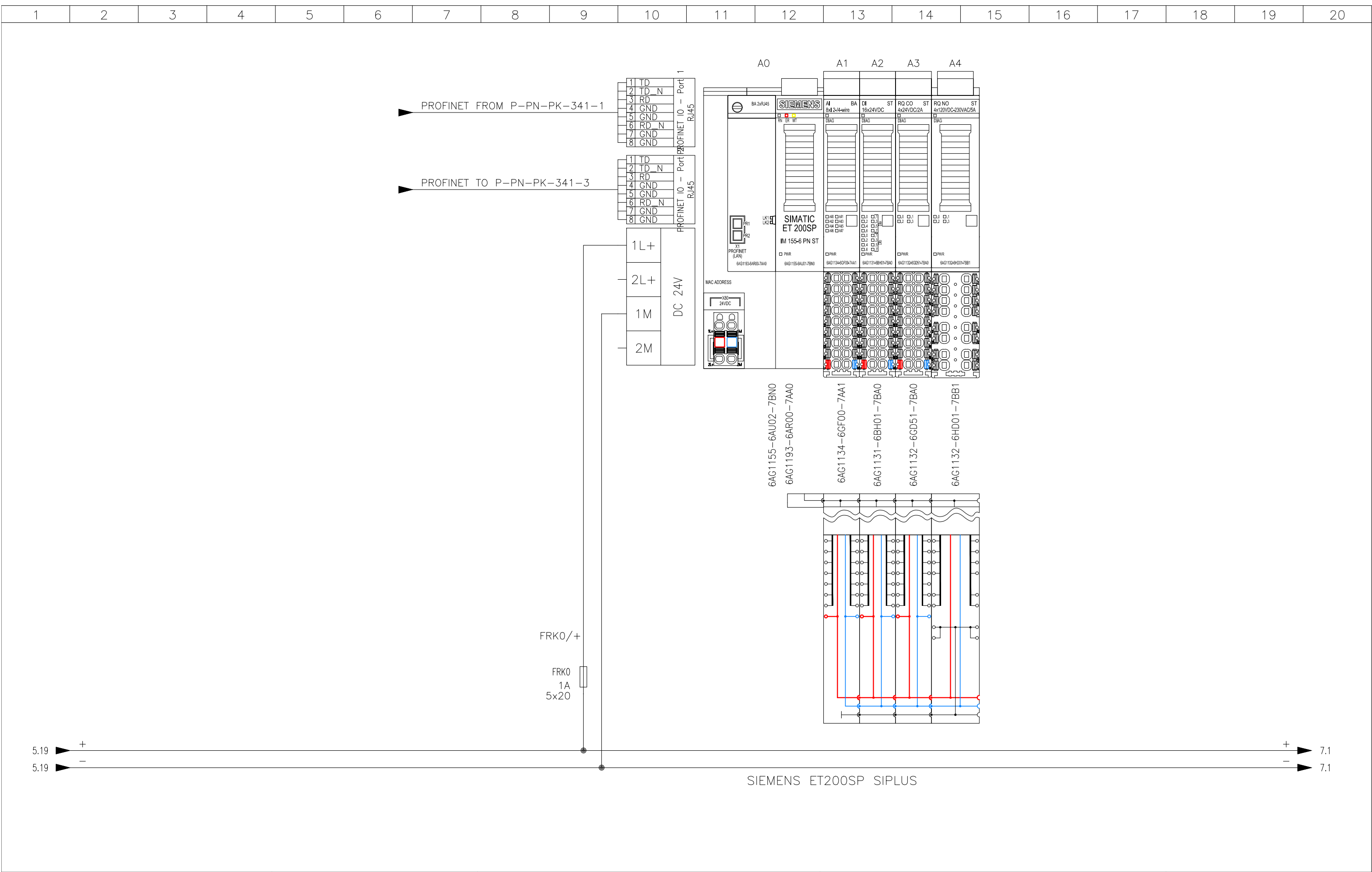


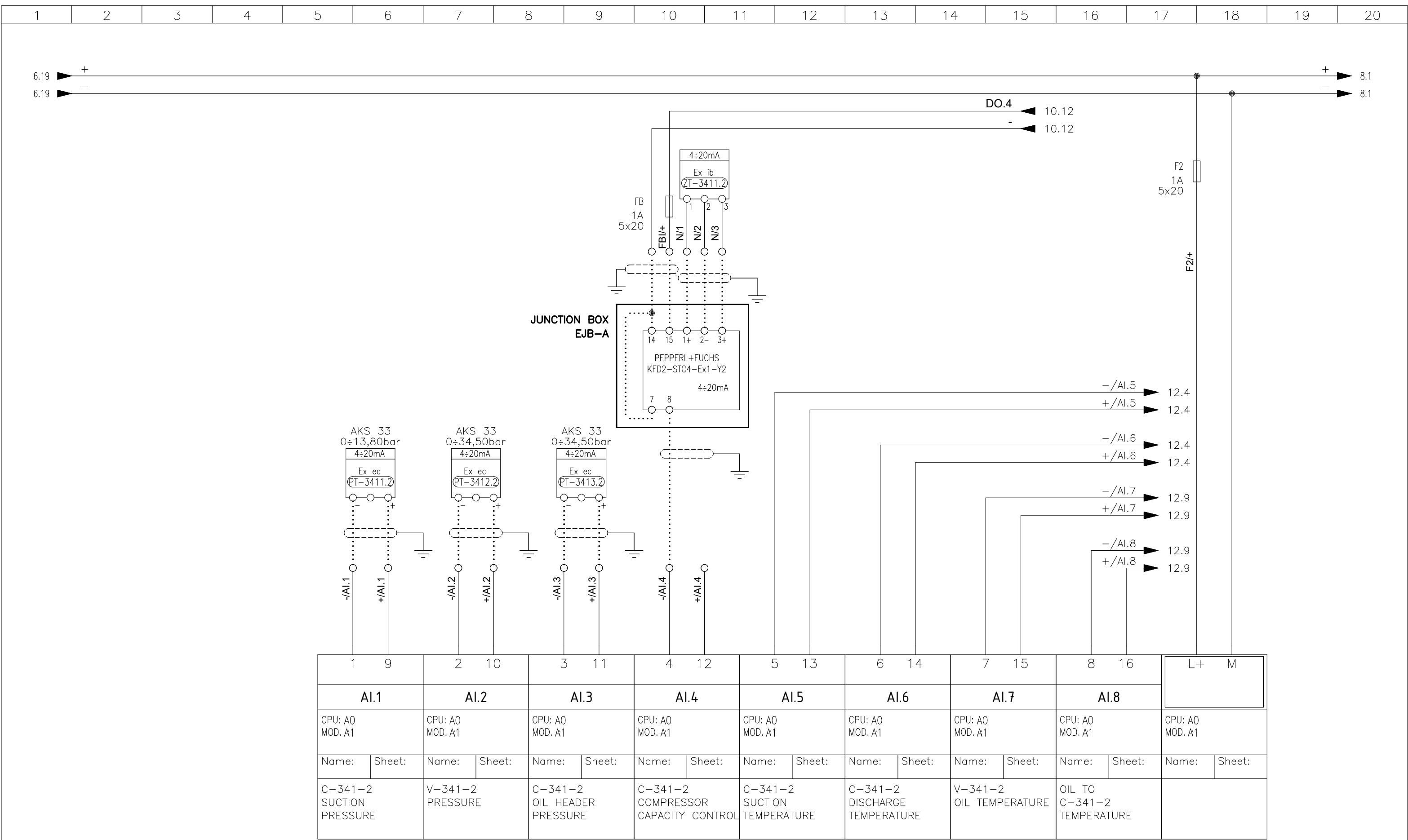
CONDUCTOR PROTECTION	TECHNOR PSRC-220																TYPE		CONDUCTOR PROTECTION
	20A																RATED BREAKING CAPACITY Icu		
																	RATED or THERMAL CURRENT In or Ith		
																	MAGNETIC RELEASE CHARACTERISTICS Im		
																	RATED R.C. SENSITIVITY Idn		
																	WIRE SECTION		
																	CABLE TYPE		
																	CABLE CARRYING LOAD Iz		
																	CABLE LENGHT		
																	LINE ID		
WIRE ID ON TERMINAL																			

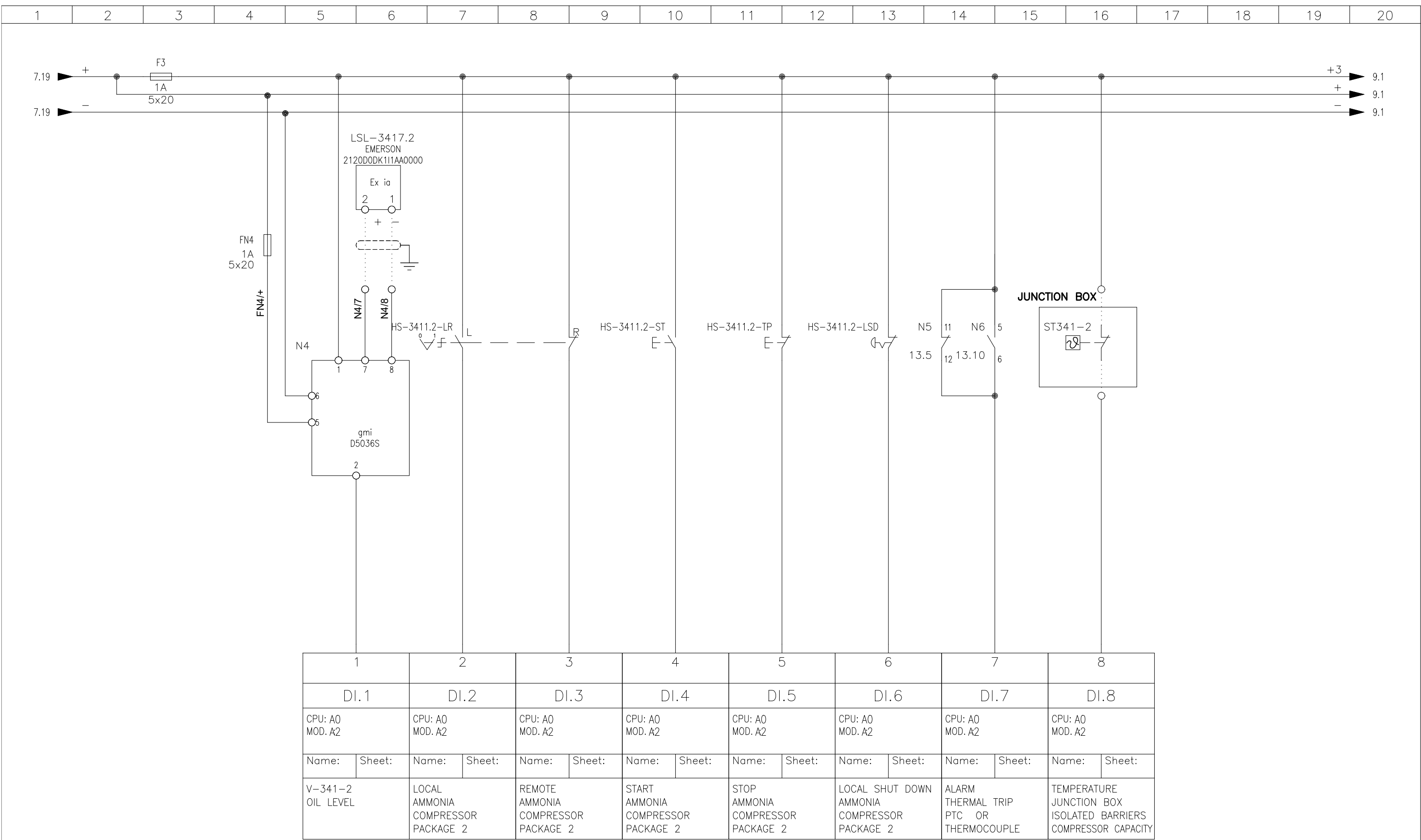


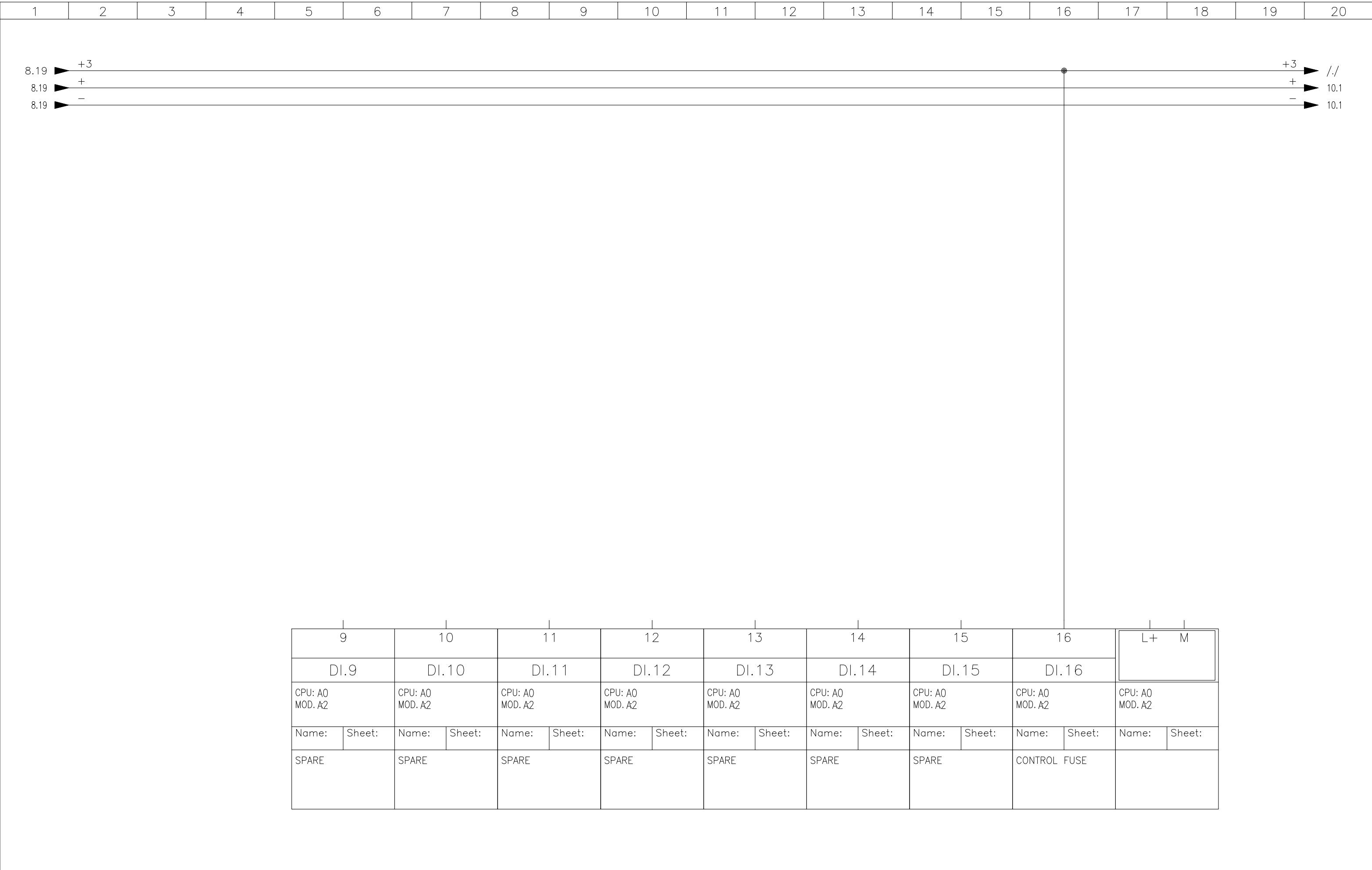
G1: TRIO-PS-2G/1AC/24DC/10 - Power supply unit PHORNIX CONTACT

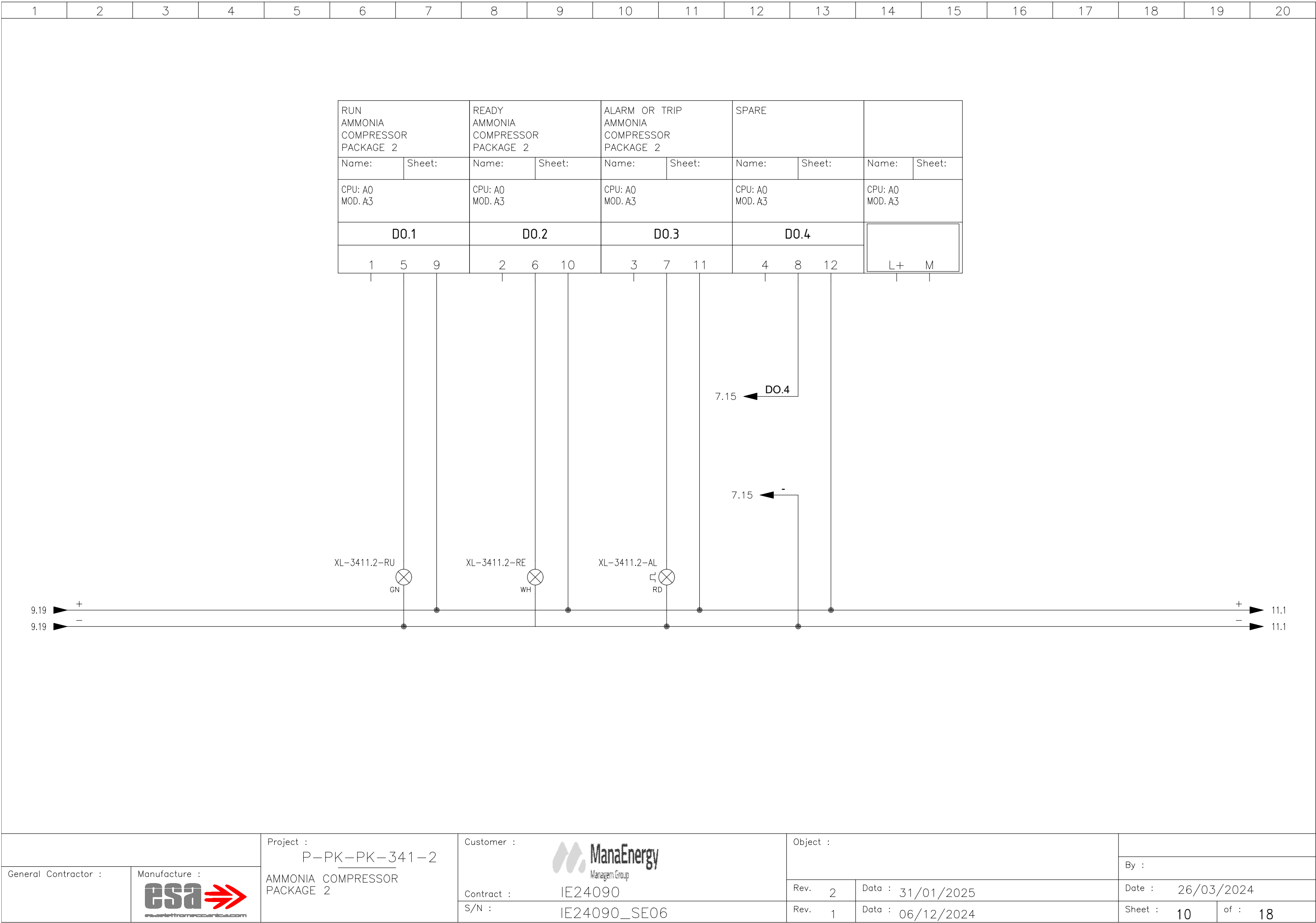
General Contractor : Manufacture : 		Project : P-PK-PK-341-2 AMMONIA COMPRESSOR PACKAGE 2	Customer : Contract : IE24090 S/N : IE24090_SE06		Object : Rev. 2 Data : 31/01/2025 Rev. 1 Data : 06/12/2024		By :	
							Date : 26/03/2024	Sheet : 5 of : 18

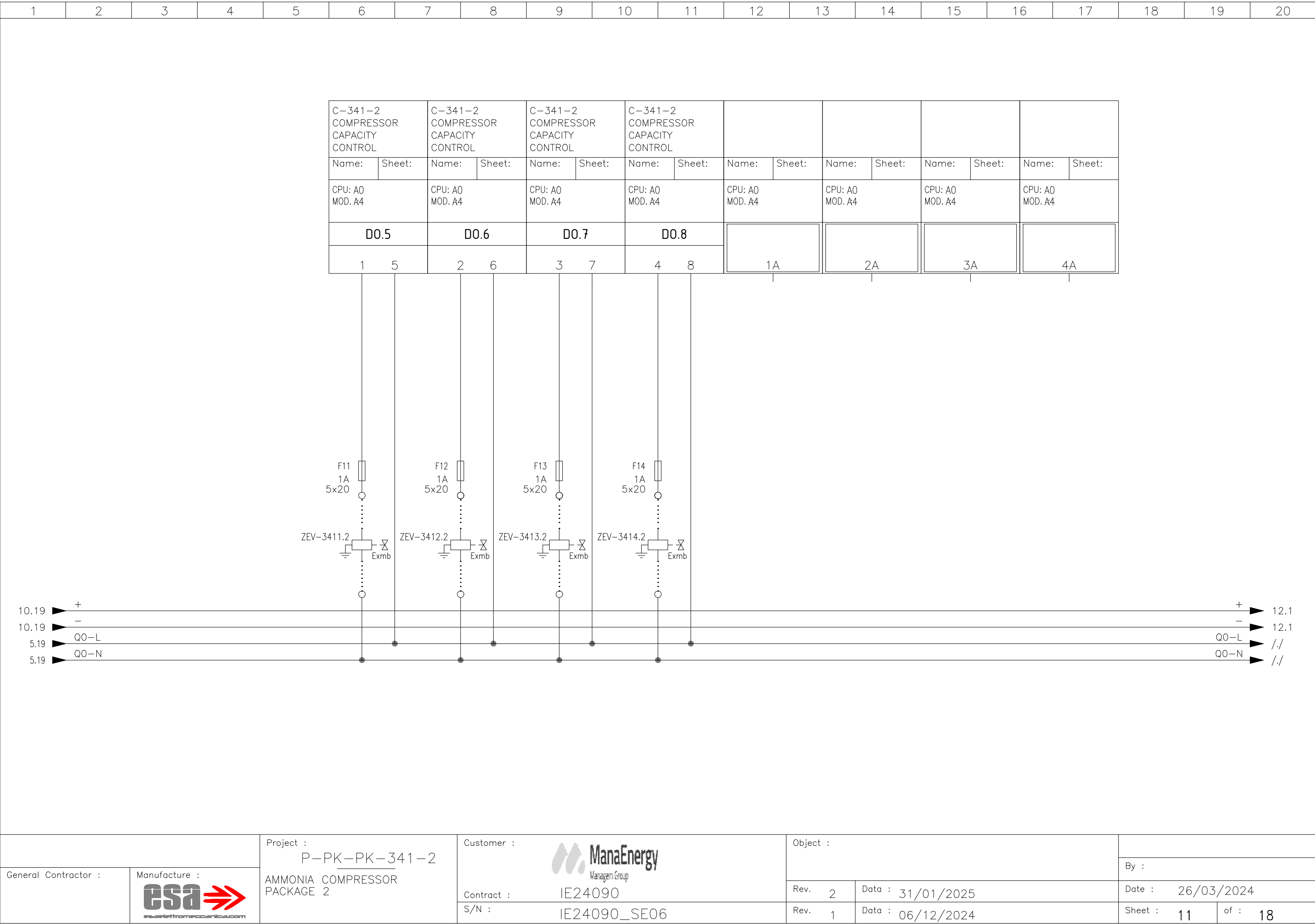


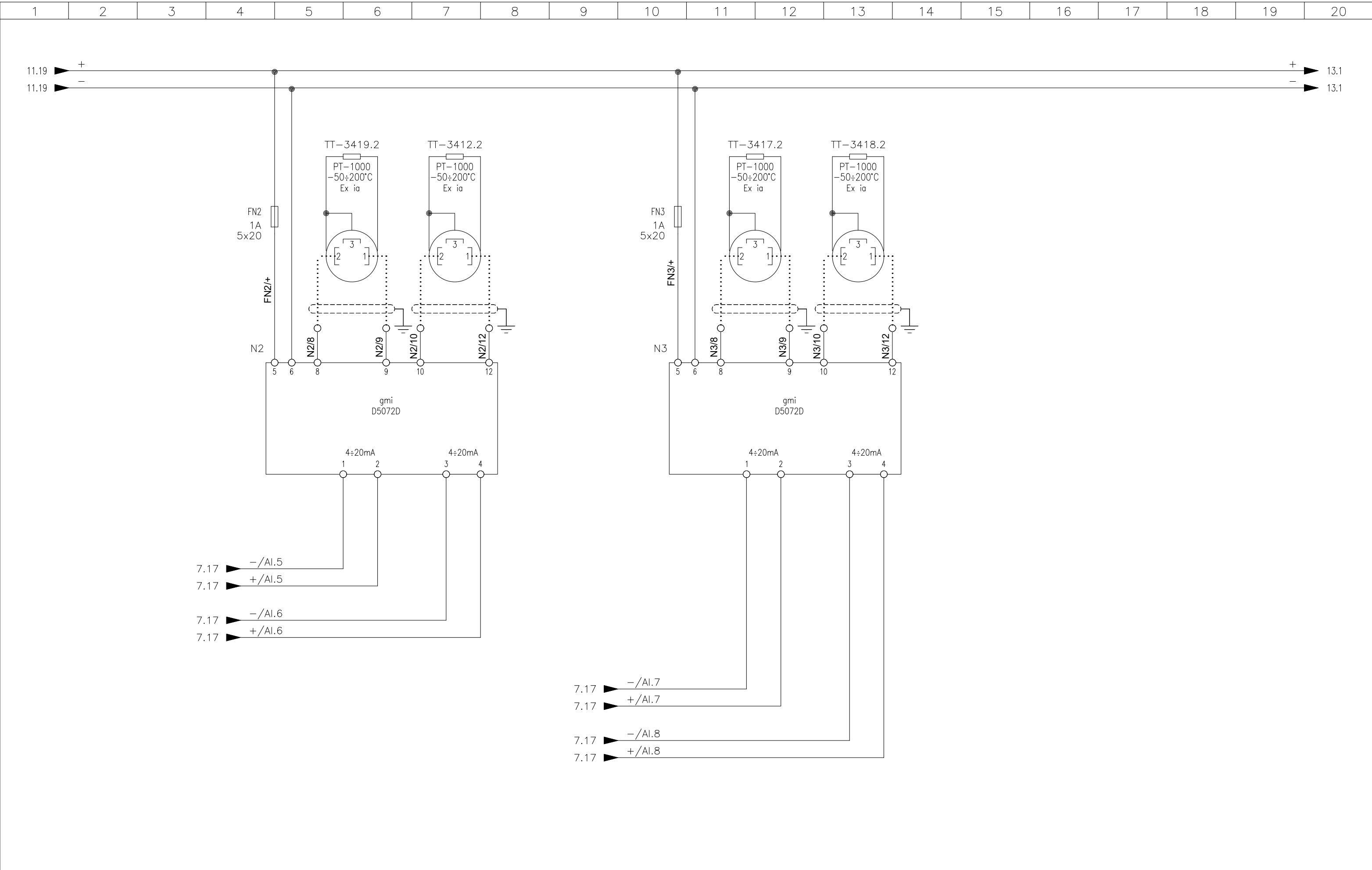


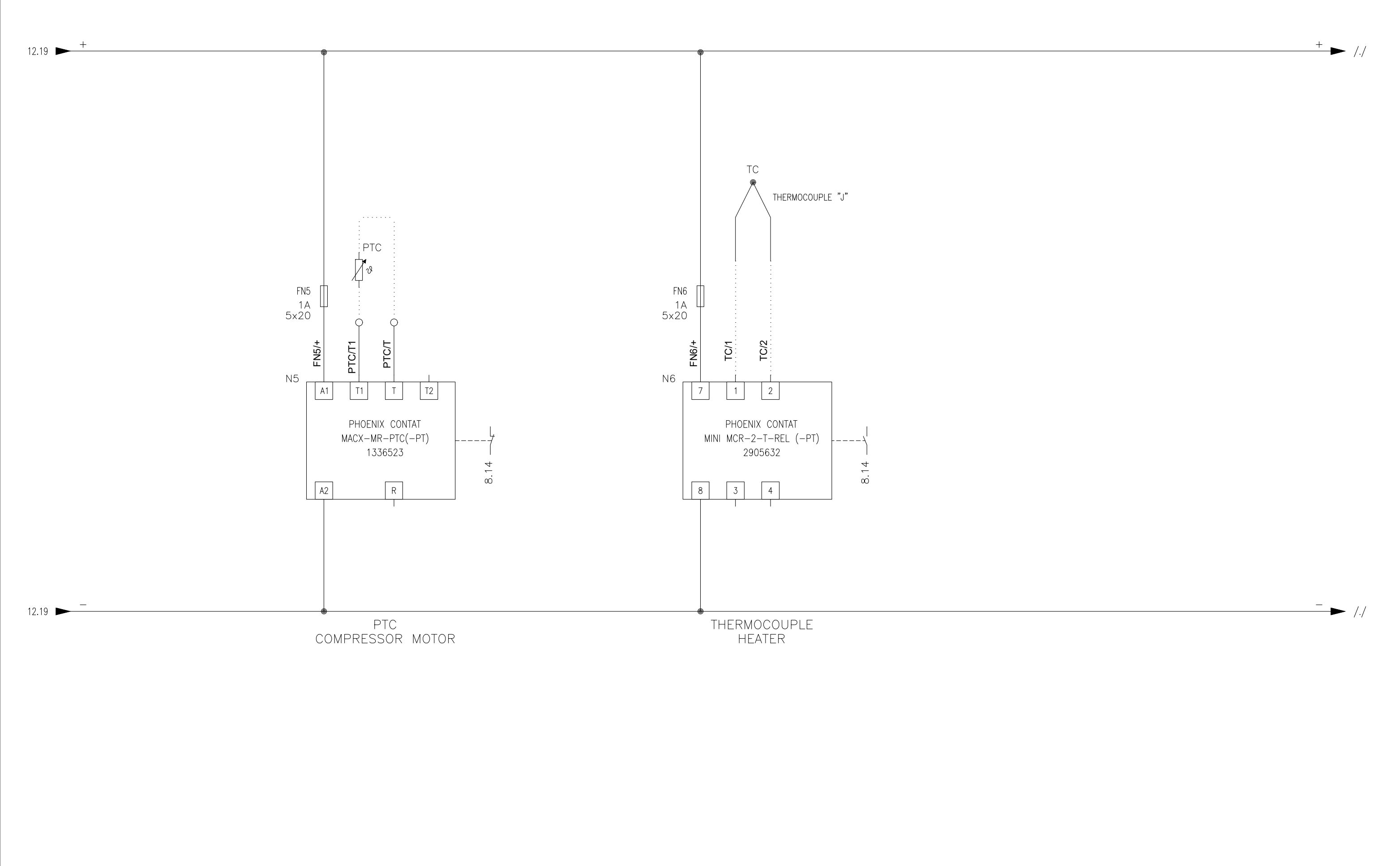


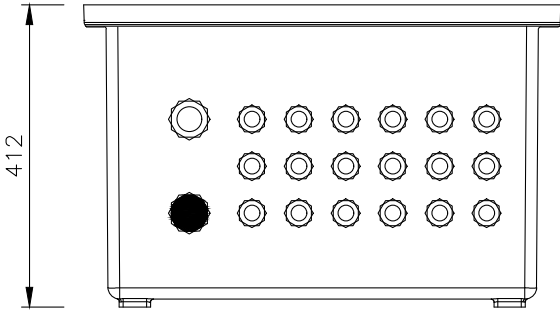
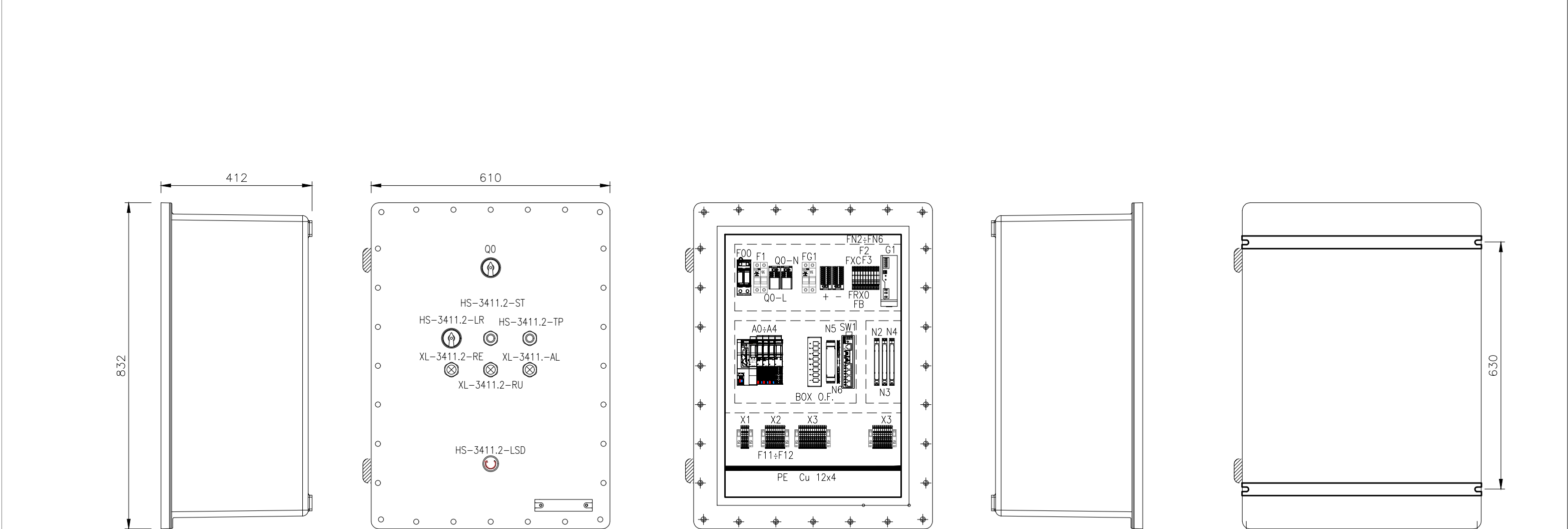












C.M.C.
Antideflagranti s.r.l.

MILANO-ITALY Tel 024455358 Fax 0248401741

Type: EJB-H

230Vac/50Hz 24Vdc 10A

Max Dis. P: 81 W

Year s.n 25/CMC/0006



0722



II 2GD

INERIS18ATEX0026X/00

Ex db [ia Ga] IIB T6

Ex tb [ia Da] IIIC T85 °C IP66

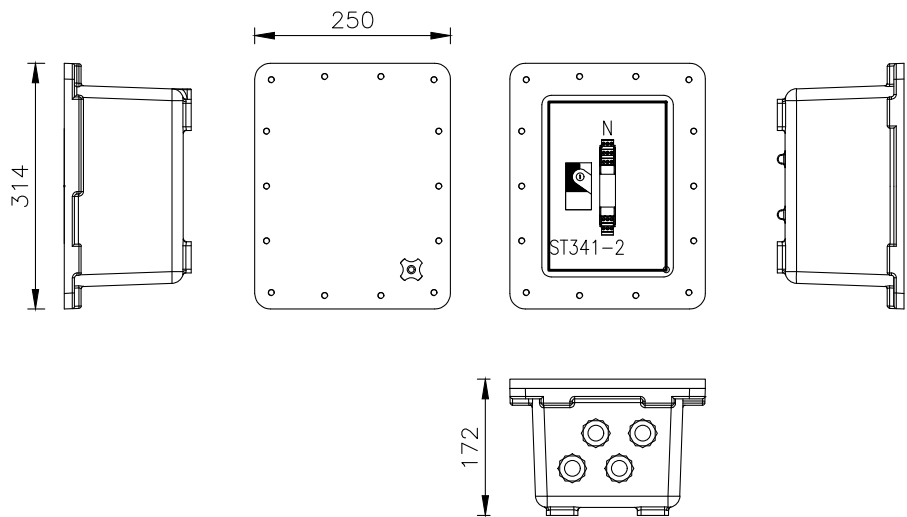
-20°C < Tamb < +50°C

Cable entries: see instructions

WARNINGS: DO NOT OPEN WHILE ENERGIZED
DO NOT OPEN IF AN EXPLOSIVE ATMOSPHERE MAY BE PRESENT
READ INSTRUCTIONS NOTE BEFORE INSTALLATION AND USE

General Contractor : Manufacture : 		Project : P-PK-PK-341-2 AMMONIA COMPRESSOR PACKAGE 2	Customer :  Contract : IE24090 S/N : IE24090_SE06		Object : Rev. 2 Data : 31/01/2025 Rev. 1 Data : 06/12/2024		By :		Date : 26/03/2024	
							Sheet : 14		of : 18	

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20							
INITIALS	SUPPLIER PN		DESCRIPTION															SUPPLIER								
–	3273110		PTFIX 6/18X2,5–NS35 GY – Distribution block															PHOENIX CONTACT								
+	3273110		PTFIX 6/18X2,5–NS35 GY – Distribution block															PHOENIX CONTACT								
A0	6AG1155–6AU02–7BN0		ET 200SP, PROFINET interface module IM 155–6PN Standard															SIEMENS								
A0	6AG1193–6AR00–7AA0		ET 200SP BA 2xRJ45															SIEMENS								
A1	6AG1134–6GF00–7AA1		AI 8xl 2–/4–wire BA															SIEMENS								
A1	6AG1193–6BP00–7DA0		ET 200SP BU15–P16+A0+2D push–in terminals															SIEMENS								
A2	6AG1131–6BH01–7BA0		ET 200SP DI 16x24 V DC															SIEMENS								
A2	6AG1193–6BP00–7BA0		ET 200SP BU15–P16+A0+2B push–in terminals															SIEMENS								
A3	6AG1132–6GD51–7BA0		ET 200SP RQ 4x24VDC/2A CO ST															SIEMENS								
A3	6AG1193–6BP00–7BA0		ET 200SP BU15–P16+A0+2B push–in terminals															SIEMENS								
A4	6AG1132–6HD01–7BB1		ET 200SP RQ 4x 120V DC..230VAC/5A ST															SIEMENS								
A4	6AG1193–6BP20–7BB0		ET 200SP BU20–P12+A4+0B push–in terminals															SIEMENS								
F00	952110		DG M TT 2P 275 – SPD															DEHN								
F1	2301138		Fuseholder for cylindrical fuses – 1P+N for fuses 10,3x38mm – BCH 1x38N 32A 690V															ITALWEBER								
F11	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE								
F12	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE								
F13	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE								
F14	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE								
F2	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE								
F3	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE								
FB	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE								
FG1	2302038		Fuseholder for cylindrical fuses – 2P for fuses 10,3x38mm – BCH 2x38 32A 690V															ITALWEBER								
FN2	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE								
FN3	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE								
FN4	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE								
FN5	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE								
FN6	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE								
FRK0	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE								
FXC	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE								
G1	2903149		TRIO–PS–2G/1AC/24DC/10 – Power supply unit															PHOENIX CONTACT								
HS-3411.2-LR	PSRC–D220		CHANGE OVER SWITCH															ITALWEBER								
HS-3411.2-LS0	PLC–S		MUSHROOM HEAD MAINTAINED (TWIST TO RELEASE)															CMC								
HS-3411.2-LS0	E20GE11		CONTACT BLOCK N.O. +N.C. WITH FLANGE															ELFI								
HS-3411.2ST	PLA30		PUSH BUTTON GREEN															CMC								
HS-3411.2ST	E20GE11		CONTACT BLOCK N.O. +N.C. WITH FLANGE															ELFI								
HS-3411.2TP	PLA20		PUSH BUTTON RED															CMC								
HS-3411.2TP	E20GE11		CONTACT BLOCK N.O. +N.C. WITH FLANGE															ELFI								
N2	D5072D		I.S. SIL2 Multifunction Temperature Converter															GMI								
N3	D5072D		I.S. SIL2 Multifunction Temperature Converter															GMI								
N4	D5036S		I.S. SIL2 Switch/Proximity Relay–Out Repeater															GMI								
N5	1336523		MACX–MR–PTC – Monitoring relay															PHOENIX CONTACT								
N6	2905632		MINI MCR–2–T–REL – Limit value switch															PHOENIX CONTACT								
P-PK-341-2	EJB–H		ENCLOSURE 832x612x412 mm L x H x D															CMC								
Q0	PSRC–320		SELECTOR SWITCH															CMC								
Q0–L	3273066		PTFIX 6/6X2,5–NS35 GY – Distribution block															PHOENIX CONTACT								
Q0–N	3273066		PTFIX 6/6X2,5–NS35 GY – Distribution block															PHOENIX CONTACT								
XL-3411.2	PLD30		PILOT LAMP GREEN															CMC								
XL-3411.2	020PD9		LAMPHOLDER BA9S, DIRECT SUPPLY															ELFI								
XL-3411.2-AL	PLD20		PILOT LAMP RED															CMC								
XL-3411.2-AL	020PD9		LAMPHOLDER BA9S, DIRECT SUPPLY															ELFI								
XL-3411.2-RE	PLD10		PILOT LAMP WHITE															CMC								
XL-3411.2-RE	020PD9		LAMPHOLDER BA9S, DIRECT SUPPLY															ELFI								
General Contractor : Manufacture : 			Project : P–PK–PK–341–2 AMMONIA COMPRESSOR PACKAGE 2					Customer :  Contract : IE24090 S/N : IE24090_SE06					Object :													
													Rev. 2				Data : 31/01/2025				Date : 26/03/2024					
													Rev. 1				Data : 06/12/2024				Sheet : 15			of : 18		



INITIALS	SUPPLIER PN	DESCRIPTION	SUPPLIER
EJB-A	EJB-A	ENCLOSURE 250x314x172 mm L x H x D	CMC
N	KFD2-STC4-Ex1-Y2	SMART Transmitter Power Supply	PEPPERL+FUCHS
ST341-2	TMS NC	THERMOSTAT	O.ERR



MILANO-ITALY Tel 024455358 Fax 0248401741

Type: EJB-A
24Vdc 10A
Max Dis. P: 6 W
Thermal probe: 55±5 °C
Year s.n 25/CMC/0012

CE 0722

II 2GD

INERIS18ATEX0026X/00

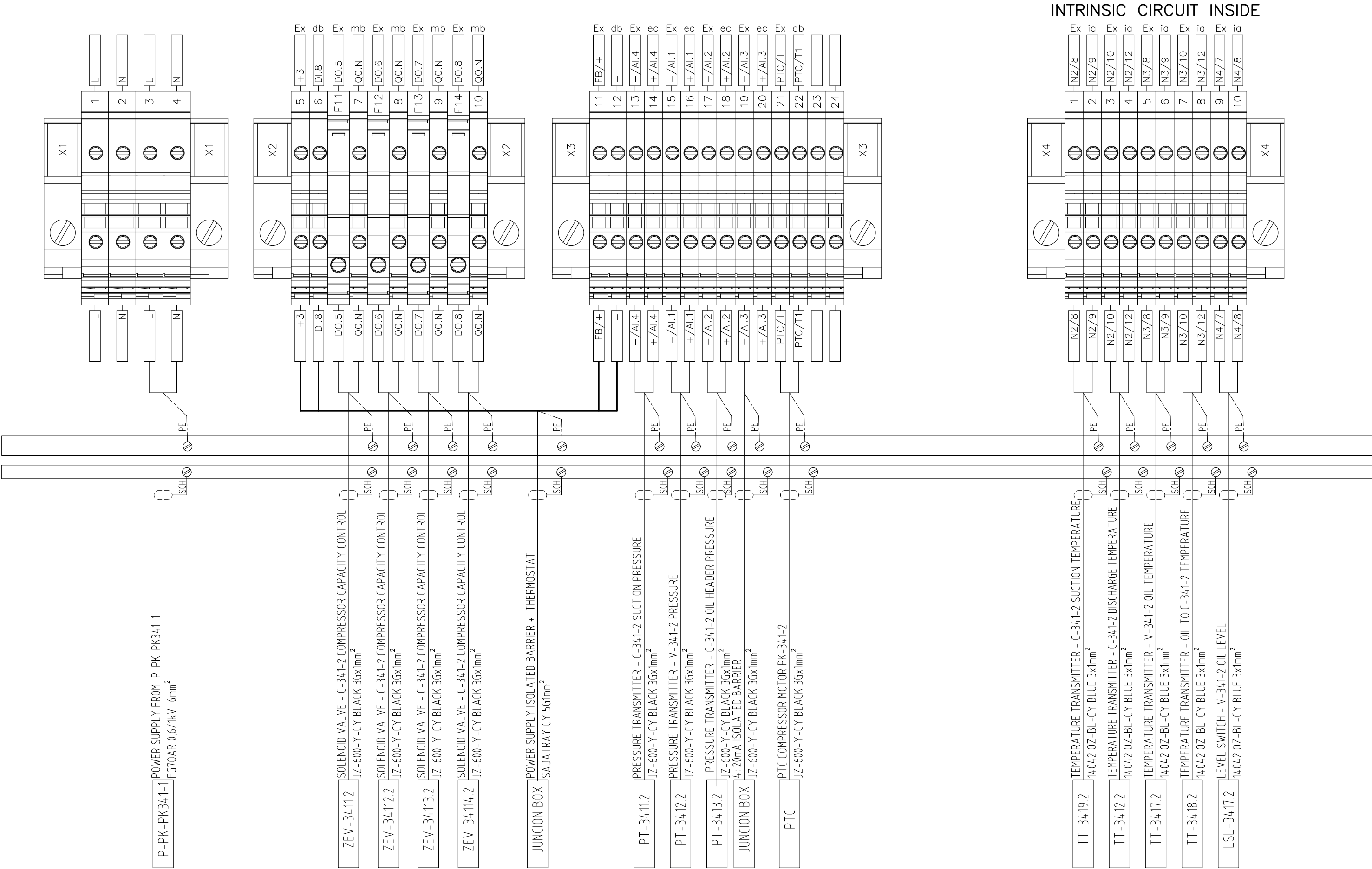
Ex db [ia Ga] IIB T6

Ex tb [ia Da] IIIC T85 °C IP66

-20°C < Tamb < +50°C

Cable entries: see instructions

WARNINGS: DO NOT OPEN WHILE ENERGIZED
DO NOT OPEN IF AN EXPLOSIVE ATMOSPHERE MAY BE PRESENT
READ INSTRUCTIONS NOTE BEFORE INSTALLATION AND USE



INTRINSIC CIRCUIT INSIDE

TYPE OF PROTECTION	
FLAMEPROF ENCLOSURE	EX db
INCREASED SAFETY	EX ec
INTRINSIC SAFETY	EX ia
ENCAPSULATRION	EX mb